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Programme: SE / CS / DS / TM	

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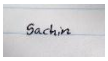
Signed on behalf of the group: 

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Project Proposal

EventiFy

An Intelligent Web and Mobile Platform for Automated Event Planning and Coordination

Developed by,
Group 88
Plymouth Batch 13
PUSL2021

Chapter 01:

Problem Statement

Event planning, especially for significant events such as weddings, corporate events, or festivals, poses several challenges. Individuals and organizations often find it hard to be in communication with more than one service provider, schedule their services, and finally, to make sure that the event is happening according to their expectations. Clients come to Sri Lanka from abroad, and even Sri Lankans themselves, sometimes find it hard to find a reliable vendor for their venue, catering, decor, lighting, music, and other services, etc. There are also no reliable platforms to find it valuable information online or make bookings. Having no central website, phone app, or other application leads to inefficiencies, miscommunication, double bookings, and wasted expenses. Planning out an event or function can take hundreds of hours, by having to make phone calls or find information online, or online reviews, and it's a not a very reliable process. As it stands now, a good portion of event planning is manual and not automatic. A lot of event organizers, and clients, are using manual processes or making a phone call or chasing through the Internet to find apps or a reliable resource. So this research intention is to form a centralized and digital process to provide a unified web and mobile application that allows a user to easily plan their function and service providers simultaneously to effectively manage their real-time bookings and service distributed online, respectfully.

Measuring the problem is done by looking into the occurrences of scheduling conflicts, slow communication, and customer dissatisfaction which are registered in the event plan weddings or events, event managers and suppliers like hotels, DJs, caterers, and light companies are among the users impacted the most by this issue. A central, AI supported event management system will be established in the project to decrease the so called operational inefficiencies, avoid double bookings and enhance the satisfaction of both users and vendors at the end of it all

Chapter 02:

Project Description

Background and Rationale

As the need for a well-planned and flawless experience increases, the professional and inherent need for technology to support the process of planning becomes evident. Event planners in Sri Lanka will often deal with juggling multiple service providers to ensure that the designated planning with the client mirrors what is needed in all aspects of an event. Current solutions acknowledge planning, but either do so in a segmented format, or it lacks the ability to be user friendly in recommending service. An artificial intelligence-powered

Project Objectives and Explanations

1. Centralized web and mobile event management system for vendor selection, event planning, and personalization of features for the event.

2. Include AI-based suggestions for preferred lodging, DJs, lighting, and decor based on the users' preference and budget.

3. A place for messaging and chatting between the users and vendors.

4. A calendar-based booking view will let the vendor manage availability, schedules, and resources.

5. Allow for online payments in a secure environment; partial deposits are accepted and confirmed.

6. Generate receipts and reports digitally for record-keeping and transparency purposes.

Project Keywords :

- Event Management
- AI Recommendation
- Vendors management

Booking System Scope :

The following are included: User registration and profile management; Event creation and customization; AI-powered recommendations for event planners; - Service listing and vendor registration. Users and vendors can chat or message each other. Scheduling of vendors on a calendar with online and partial payments integrated; Notification via SMS and email - Admin dashboard with Vendor and Event tracking Excluded: Physical execution of the event; Manual vendor verification after initial registration; Offline processing of payments upon deposit

Chapter 03
Research Gap

3.1 Literature Review Comparison

A number of previous studies have looked into event management systems, but most of them don't have the level of AI integration, live communication, or vendor scheduling that our project does. By looking at these side by side, we can see where the AI-Powered Event Management System (AI-EMS) wants to fill in the gaps.

Razali and colleagues (2023) examined Event Management Systems (EMS) focusing on registration, scheduling modules, and attendee data management through a web-based platform. Their service improved administrative workflows, however, it did not offer features based upon recommendations or predictive services. In addition, it did not offer a capability for live vendor matching or automating the planning process with AI; our system builds on this by offering an AI Event Planner that recommends optimized bundles of services based upon skills/behaviors of the client.

Patel et al. 2019 in IRJET designed a web-based system whereby the aspects of event registration, participant management, and venue booking are digitalized. The system, although usable, did not have any real-time or cloud-based functions, dynamic chat capability, or an automated vendor coordination system. Our system goes a step beyond by integrating real-time chat and vendor calendar booking capabilities for communication and scheduling operations.

Perera (2014) demonstrated that various aspects of an academic environment could be manipulated from a single source by studying event systems at a university. The event system was not mobile and did not involve AI in its design and only addressed events that existed on campus. Our project facilitates the use of commercial events in a non-academic setting, and provides a cross platform to inform smart decisions.

Balute et al. (2017) did not consider user-based event coordination. Instead, they studied event infrastructure systems and IT infrastructure in monitoring the associated systems with a focus on performance metrics and the monitoring of resources for events. One of the primary design considerations was the trustworthiness of the system.

Additionally, the paper published on HAL (HAL-04344837) presents a set of sophisticated architectures for event scheduling and resource allocation via optimization algorithms in large-scale systems. This paper focuses on automating the matching of locations, time slots, and resource constraints. This research does not use AI-based inference of user preferences or discuss any real-time vendor chat. Although HAL provided a very solid scheduling model, AI-EMS builds upon that work with the addition of user-focused AI (recommendations, adaptability) and communication integration to create a more interactive and personalized experience for event planning.

In short, none of these four existing systems or HAL architectures provide a fully integrated AI + communications + vendor scheduling platform for common event management processes. The technology gap lies in the convergence of AI recommendations, real-time chat, calendar booking, context-awareness and context-sensitivity (or same-day event adaptation) into a single fully integrated user experience platform - especially in real-world environments such as Sri Lanka.AI-EMS fills these gaps, providing an intelligent, scalable, and communication capabilities, a platform that pushes academic and practical work in new directions.

Chapter 04:
Requirements Analysis

Functional Requirements

1. Ability for users to log in and manage their profiles.
2. Turning events on, personalization, and picking add-ons.
3. Al-powered suggestion system for locations, music, and lighting.
4. Vendor sign-up and management of service listings.
5. Booking system based on calendar for vendors.
6. Payment with a 20% deposit and confirmation.
7. Chatting/messaging system between users and vendors.
8. Notification system (letters and SMS).
9. Admin dashboard for tracking and reporting.

Non-Functional Requirements

1. The system must be capable of supporting 500 users simultaneously.
2. Both the web and the mobile applications should be ready for use within 5 seconds.
3. The use of encryption for protection of sensitive data.
4. A scalable design that can support future Al feature developments.
5. Interfaces that are easy to use and have responsive design.

User Requirements

• Users must have the capability of setting up events and choosing vendors without difficulties.

• Vendors ought to be able to handle their availability and services without any technical help.

• Admins are supposed to keep an eye on system performance and event progress.

Methods to Collect Requirements

• Focus groups including locatable users and vendors.
• Online questionnaires and interviews.
• Research on the event management platforms already existing.
• Study of the literature related to Al recommendation systems in event planning

Chapter 05: Finance

Item	Description	Estimated Cost (LKR)
Domain & Hosting	web hosting and domain registration	30,000
SSL Certificate	Secure Sockets Layer (SSL) Certificate	5,000
Email & Notification Services	SMTP service or Firebase Cloud Messaging setup	10,000

Financial Justification: By lowering planning errors, boosting event efficiency, enhancing user experience, and opening the door for scalable AI-powered solutions, the estimated LKR 30,000 total cost is justified.

Chapter 06: External Organizations

- Tech Lanka (Pvt) Ltd: To deployment of prototypes
- Sri Lanka Chamber of Commerce: Promote vendor cooperation and test-runs.
- Local event vendors: Take part in system testing and offer usability comments.

Chapter 07: Time Frame / Timeline

Phase	Duration	Deliverable
Requirement Gathering	4 Week	Requirement Document
Design	4 Week	UML Diagrams, Wireframes
Development	14 Week	Working Prototype (Web + Mobile)
Testing	5 Week	Test Report, Bug Fixes
Documentation	4 Week	Final Feasibility Report and Submission

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